



Automatic Identification of Missing Persons in Search and Rescue Operations using Unmanned Aerial Vehicles

Projeto de Engenharia Eletrotécnica e Computadores

Afonso Marques Rodrigues

Supervisores:

Doutor Nuno Alexandre Antunes Martins Pessanha Santos

Doutor Bruno Duarte Damas

Mestrado em Engenharia Eletrotécnica Militar

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Abstract

Every year, numerous individuals go missing, often in disaster-stricken areas, leading to tragic outcomes if they are not found in time. The integration of Unmanned Aerial Vehicles (UAVs) equipped with RGB and Infrared Thermal Cameras (ITC) has shown significant potential in enhancing efforts to locate missing individuals. This thesis explores the application of computer vision algorithms to create a model capable of automatically detecting and tracking a missing person from the perspective of a UAV. By studying and developing a new specialized dataset and leveraging state-of-the-art Deep Learning (DL) object detection algorithms such as YOLO and DETR, along with Multi-Object Tracking (MOT) algorithms like DeepSORT and ByteTrack, the study aims to enhance the efficiency and accuracy of locating individuals in critical scenarios. The proposed approach will use three data fusion frameworks based on the data types: RGB-based, ITC-based, and a fusion-based approach, using both data types to address challenges such as complex environments and the need for real-time processing in Search and Rescue (SAR) operations. This work aims to identify the most effective model for detecting individuals in real-time, significantly contributing to advancing SAR technologies.

Key Words

Computer Vision; Deep Learning; Multi-Object Tracking; Object Detection; Search and Rescue; Unmanned Aerial Vehicles.

Resumo

Todos os anos, inúmeras pessoas desaparecem, muitas vezes em zonas afetadas por catástrofes, o que pode ter consequências trágicas se não forem encontradas a tempo. A integração de Veículos Aéreos Não Tripulados (VANT) equipados com câmaras *Red, Green and Blue* (RGB) e *Infrared Thermal* (IT) têm demonstrado um potencial significativo na melhoria dos esforços de localização de indivíduos desaparecidos. Esta tese explora a aplicação de algoritmos de visão computacional para criar um modelo capaz de detetar e seguir automaticamente uma pessoa desaparecida a partir da perspetiva de um VANT. Através do estudo e desenvolvimento de um novo conjunto de dados especializado e da utilização de algoritmos de deteção de objetos de Aprendizagem Profunda (AP) de estado-da-arte, como o *YOLO* e o *DETR*, juntamente com algoritmos de Seguimento de Multi-Objetos (SMO), como o *DeepSORT* e o *ByteTrack*, o estudo visa melhorar a eficiência e a precisão da localização de indivíduos em cenários críticos. A abordagem proposta utilizará três estruturas de fusão de dados com base nos tipos de dados: Baseado em *RGB*, baseado em *IT*, e uma abordagem baseada em fusão, utilizando ambos os tipos de dados para enfrentar desafios como ambientes complexos e a necessidade de processamento em tempo real em operações de Busca e Salvamento (BS). Este trabalho tem como objetivo identificar o modelo mais eficaz para a deteção de indivíduos em tempo real, contribuindo significativamente para o avanço das tecnologias utilizadas em BS.

Palavras Chave

Visão Computacional; Aprendizagem Profunda; Seguimento de Objetos; Deteção de Objetos; Busca e Salvamento; Veículos Aéreos Não Tripulados.

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Acronyms

2D two-dimensional.

3D three-dimensional.

AP Average Precision.

BB Bounding Box.

C2 Command and Control.

CNN Convolutional Neural Networks.

COCO Common Objects in Context.

COTS Commercial-Off-The-Shelf.

CV Computer Vision.

CVAT Computer Vision Annotation Tool.

DBT Detection-Based tracking.

DeepSORT Deep Simple Online and Real-time Tracking.

DETR DETection TRansformer.

DFT Detection-Free tracking.

DJI Da-Jiang Innovations.

DL Deep Learning.

EPRU Emergency Protection and Rescue Unit.

F-RCNN Faster-Regional Convolutional Neural Networks.

FN False Negative.

FP False Positive.

fps frames per second.

GCS Ground Control Station.

GPU Graphical Processing Unit.

GT Ground Truth.

HOTA Higher Order Tracking Accuracy.

ID Identifier.

IDS Identity Switches.

IoU Intersection over Union.

IR Infrared.

ISR Intelligence, Surveillance and Reconnaissance.

ITC Infrared Thermal Camera.

KF Kalman Filter.

ML Machine Learning.

MOT Multi-Object Tracking.

MOTA Multi-Object Tracking Accuracy.

NAS Neural Architecture Search.

NII-CU National Institute of Informatics - Chiba University.

NN Neural Network.

NRG National Republican Guard.

OpenCV Open Source Computer Vision Library.

OS Open Source.

P Precision.

PMGI Proximity Maximization Gradient Integration.

R Recall.

RGB Red, Green and Blue.

RoI Region of Interest.

RPN Region Proposal Networks.

SAR Search And Rescue.

SDNet Structure-Preservation Detail Network.

SORT Simple Online and Real-time Tracking.

SOTA State-Of-The-Art.

TP True Positive.

UAV Unmanned Aerial Vehicle.

YOLO You Only Look Once.

1 Introduction

Daily reports indicate that individuals go missing in Portugal, from the northern to the southern regions. The National Republican Guard (NRG), specifically its Emergency Protection and Rescue Unit (EPRU), is a military force responsible for conducting Search And Rescue (SAR) operations [1]. This unit is composed of military specialized teams using Unmanned Aerial Vehicles (UAVs) to locate missing persons as quickly as possible to enhance their chances of survival. The location, terrain type, time of day, and other factors complicate detecting and tracking these individuals.

This thesis aims to develop a Computer Vision (CV) framework that automatically detects and tracks individuals using image data collected by UAVs to enhance and support SAR operations conducted by EPRU field teams. The development of this framework will involve close collaboration with the EPRU, which will facilitate the use of UAVs currently deployed in actual operations and the acquisition of a realistic dataset. Regarding the problem at hand, this chapter will describe the motivation (Section 1.1), the problem formulation (Section 1.2), the objectives (Section 1.3), and the existing challenges (Section 1.4).

1.1 Motivation

The value of human life is immeasurable, and our collective responsibility is to protect and enhance it through ongoing advancements in various fields, including science and technology. Security forces safeguard citizens during dangerous situations, assisting and rescuing as needed. EPRU is a specialized unit that focuses on preventing and responding to severe incidents, including rural fires, hazardous materials, floods, earthquakes, and SAR operations, as well as emergencies in aquatic environments [1].

In Portugal, approximately 4,000 individuals are reported missing each year [2]. A significant concern is the disappearance of elderly people who can suffer from mental and physical impairments. Since 2018, over 1,500 individuals aged 65 or older have been reported missing [3]. This substantial number highlights the urgent need to establish dedicated teams focused on searching for these vulnerable individuals, decreasing the search time and increasing the probability of finding the person alive and in good health. Such teams can include human personnel, canine units, or UAVs, with UAVs currently receiving the most attention. Thanks to remarkable advancements in recent

years, we now have access to UAVs equipped with advanced cameras that can autonomously cover vast areas, decreasing operating costs and increasing the Intelligence, Surveillance and Reconnaissance (ISR) capability on the field. An example of images captured during a real SAR operation can be seen in Figure 1. A preliminary analysis indicates the potential of using UAV images for ISR and, if necessary, Command and Control (C2).



Figure 1: SAR operations in semi-arid (*left*) and mountainous (*right*) areas.

1.2 Problem Formulation

Object detection and tracking are essential CV tasks [4]. They involve identifying and localizing specific objects over time in a video sequence with high accuracy. Although humans can easily detect objects in a scene, this task is complex and challenging for computers. The appearance of humans within the observed area can change significantly due to changes in lighting conditions, background, body posture, shape deformation, and camera selection.

This thesis aims to present a robust detection framework designed to accurately locate individuals in various SAR conditions using UAV camera images. We are collaborating closely with the unit responsible for SAR operations, which allows our research to consider their hardware, operational procedures, and limitations. To achieve our goals, we will utilize synchronized Red, Green and Blue (RGB) and Infrared Thermal Camera (ITC) images while implementing algorithms for person tracking. This approach will address the challenges posed by limited visibility, complex backgrounds, and the need for real-time processing. To achieve real-time processing, we will perform our processing at a Ground Control Station (GCS) to allow access to higher processing power.

A EPRU operational team consists of military personnel capable of using UAVs in several operational scenarios while also being able to carry out its standard maintenance. The team operates a Da-Jiang Innovations (DJI) Matrice 30T [5], as shown in Figure 2 *left*, capable of transmitting real-time video to a GCS for C2, similar to most existing Commercial-Off-The-Shelf (COTS) UAV systems. As shown in Figure 2 *right*, the available camera gimbal is composed of a single module equipped with RGB camera, ITC, a laser range finder, and an auxiliary light.

These teams operate in highly diverse environments and under challenging conditions until their missions are complete. These conditions pose difficulties for CV algorithms, such as the aerial perspective of UAVs concerning people's size and position, the background's complexity, and the necessity for real-time processing. Utilizing the latest techniques in the field is essential for achieving the best possible results, as will be explored in this thesis. Currently, there are only a few publicly available datasets that integrate synchronized data from RGB and ITC images, as outlined in Section 3.5. Therefore, we will need to create a novel dataset focused on locating missing individuals in SAR environments, combining RGB and ITC images.



Figure 2: DJI Matrice 30T (*left*) and DJI camera (*right*).

1.3 Objectives

The main objective of this thesis is to create a CV framework that can track individuals in outdoor settings using RGB and ITC images taken from a UAV during SAR operations. To accomplish this goal, we have established six secondary objectives:

1. **Acquisition of a New Dataset** - Due to the limited availability of

publicly accessible datasets, it is necessary to acquire RGB and ITC images during real SAR operations, as illustrated in Figure 3;



Figure 3: RGB (*left*) and ITC (*right*) images captures from real operations.

2. **State-Of-The-Art (SOTA) Review** - This review will examine advanced methods for object detection and tracking in complex environments, focusing on integrating RGB camera and ITC technologies. The findings will help identify effective model architectures for person detection in SAR environments;
3. **Dataset Preprocessing** - Preparation of the dataset for training, testing, and evaluation by synchronizing videos, selecting relevant frames, adding annotations, and applying data augmentation if necessary;
4. **Implementation and Testing of Algorithms** - Assess the effectiveness of the proposed algorithms using performance metrics focused on accuracy, precision, and computational efficiency, and compare them with SOTA methods to ensure real-time capabilities;
5. **Real-World Testing** - Evaluate the developed model in real-time during real SAR operations to assess its effectiveness and robustness in the real world. This step will validate its performance and guide necessary adjustments;
6. **Proposed Architecture Optimization** - If necessary during the developed work, optimize the architecture to reduce parameters while maintaining high performance. This will enable deployment on devices with limited processing capacity, such as onboard UAVs processing, for real-time use.

1.4 Challenges

The achievement of the objectives described in Section 1.3 involves several challenges that must be addressed. Several research studies have been conducted on detecting missing persons in SAR environments [6, 7, 8], and some of the challenges identified have already been addressed or minimized.

A common issue with publicly available datasets is their limited scope, primarily featuring images of standing individuals. This creates a significant gap in the representation of people sitting or lying down, which is crucial for disappearance cases. Furthermore, few datasets explore ITC images in the SAR context [6, 9], and even fewer combine RGB and ITC images [10, 11]. Although integrating these two camera types in a single UAV can provide synchronized gimbal movements, it also introduces challenges such as misalignment, differences in resolution, varying frame rates, and distortion of captured images that must be addressed.

The complexity of the environment is also a significant concern. As shown in Figure 4, individuals often occupy dense vegetation and uneven terrain, making navigation difficult. Additionally, the altitude of the UAV affects detection, as individuals appear smaller from higher altitudes. Limited camera resolution and algorithms not trained on aerial images also reduce the probability of detection.



Figure 4: Examples of ITC (*left*) and RGB (*right*) images.

A critical concern is the computational cost of the algorithms. While Deep Learning (DL) algorithms, often superior, they require more processing time than traditional Machine Learning (ML) methods [12]. This raises concerns about computational cost and the feasibility of real-time detection, which must be adequately explored.

2 Background

Humans possess exceptional visual acuity, allowing adequate perception of their environment. This talent enables quick interpretation of images, pattern recognition, and distinguishing shapes and faces. Despite advancements in CV, computers struggle with identifying, tracking, and classifying images compared to humans due to the complexity of the real world and variability in factors like lighting, occlusion, and object appearances.

This chapter will explore the background theory about the camera model (Section 2.1), object detection (Section 2.2), object tracking (Section 2.3), image fusion (Section 2.4), and performance evaluation (Section 2.5).

2.1 Camera Model

An image is a two-dimensional array of pixels in a color space, and the pixel values collectively represent information [13]. To obtain meaningful insights from the captured images, describing the projection of the real world onto an image plane is essential. When using a digital camera, three-dimensional (3D) world points $\mathbf{p}_w$ are transformed into two-dimensional (2D) image coordinates $\tilde{\mathbf{x}}_s$, as described by [14]:

$$\lambda \underbrace{\begin{bmatrix} u \\ v \\ 1 \end{bmatrix}}_{\tilde{\mathbf{x}}_s} = \underbrace{\begin{bmatrix} f_y & s & c_x \\ 0 & f_x & c_y \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\mathbf{P}} \underbrace{\begin{bmatrix} \mathbf{R} & \mathbf{T} \\ 0 & 1 \end{bmatrix}}_{\mathbf{P}_w} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\mathbf{p}_w} \quad (1)$$

where λ is a scaling factor different from zero, $(u, v) \in \mathbb{R}^2$ represent the pixel coordinates, $(X, Y, Z) \in \mathbb{R}^3$ represent the 3D coordinates, $\mathbf{P}$ is the 3×4 camera matrix, $\mathbf{K}$ is the 3×3 calibration matrix containing the camera's intrinsic parameters, $\mathbf{R}$ is a 3×3 rotation matrix, and $\mathbf{T}$ is a 3×1 translation vector. The pair $\{\mathbf{R}, \mathbf{T}\}$ is called the extrinsic parameters. Regarding the intrinsic parameters, independent focal lengths f_x and f_y are considered for the sensor's x and y dimensions. The parameter s encodes any possible skew between the sensor axes due to the sensor not being mounted perpendicular to the optical axis. However, in this analysis, potential skew between the two axes on the image plane is disregarded, as current solid-state manufacturing

techniques render this effect negligible [15]. Finally, (c_x, c_y) denotes the image center expressed in pixel coordinates, as illustrated in Figure 5.

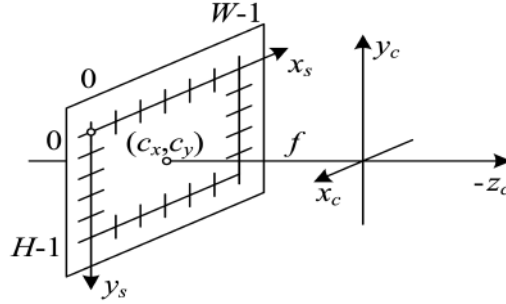


Figure 5: Simplified illustration of the camera's intrinsic parameters [15].

Most digital cameras exhibit substantial image distortion, particularly in the wide-angle lens model used in the considered UAV system. A distorted image can typically be identified by the fact that a straight line in the real world appears curved in the image representation. The most relevant lens distortions are radial and tangential, with the latter typically being much smaller in magnitude [16]. The lens distortion can be expressed as [15]:

$$\begin{aligned}
 x_{distortion} &= x (1 + K_1 r^2 + K_2 r^4) + 2p_1 xy + p_2 (r^2 + 2x^2) \\
 y_{distortion} &= y (1 + K_1 r^2 + K_2 r^4) + 2p_2 xy + p_1 (r^2 + 2y^2)
 \end{aligned} \tag{2}$$

where (x, y) are the coordinates of the undistorted points, $r^2 = x^2 + y^2$, $\{K_1, K_2\}$ are constants in the range $[-1, 1]$, and $\{p_1, p_2\}$ are constants in the range $[-0.1, 0.1]$. Typically, the intrinsic matrix $\mathbf{K}$ and the distortion coefficients $\{K_1, K_2, p_1, p_2\}$ are obtained offline with a calibration pattern [17].

2.2 Object Detection

Object detectors are designed to identify and classify objects within a single image. The objects are typically labeled with a Bounding Box (BB) and a defined classification confidence level. Object detection models can be broadly classified into one-stage and two-stage detectors [18], each tailored to balance speed and accuracy depending on the application. As Figure 6 illustrates, two-stage detectors involve a sequential pipeline generating the region proposal and performing classification.

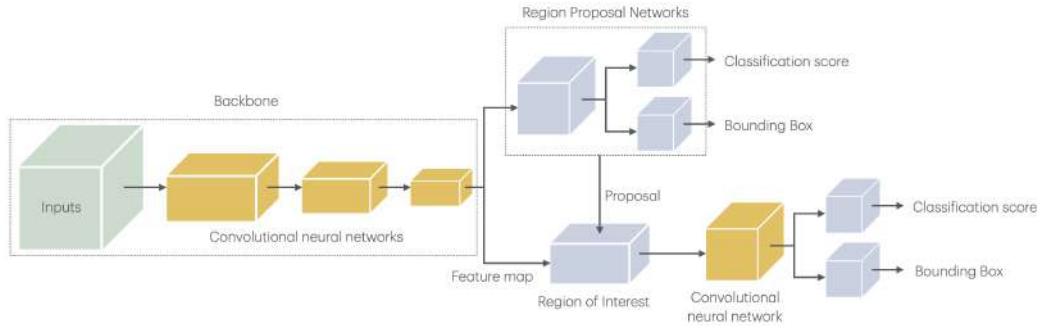


Figure 6: Basic architecture of a two-stage detector adapted from [19].

In the region proposal stage, Region Proposal Networks (RPN) proposes regions where the object could be located. In the object classification stage, a Region of Interest (RoI) pooling layer merges the proposed region with the features generated in the backbone. Subsequently, a convolutional layer is employed to classify the object and estimate a BB that indicates its location, thereby enhancing detection accuracy.

In contrast to two-stage detectors, one-stage detectors directly predict BBs using a predetermined set of regions (anchors), eliminating the requirement for using a RPN. The RoI pooling layer utilizes the features extracted from the backbone network within each of these anchors to identify and classify the object and its BB. This approach increases detection speeds, making one-stage detectors more feasible for real-time use.

The classification for object detection in images usually uses features generated by a backbone network in both architectures [4] using Convolutional Neural Networks (CNNs). This network takes an image input and generates a feature map, a simplified representation highlighting essential patterns such as edges and texture shapes. To enhance detection accuracy, deep and more densely connected backbones are used.

2.3 Object Tracking

Some models expand the object detection capability to enable object tracking, allowing the algorithm to utilize information between frames. Multi-Object Tracking (MOT) is a crucial CV task that involves detecting multiple objects and performing tracking of their trajectories across a sequence of images. Unlike object detection, which outputs a collection of rectangular BB identified by their coordinates, taking into account usually

a single frame, MOT algorithms assign a target Identifier (ID) to each box to facilitate object or class differentiation [20]. So far, about 70% of MOT research focuses on pedestrian tracking [21].

Unlike object detection, categorizing the diverse types of MOT algorithms into distinct classes presents a challenge. Based on the classification proposed in [21], we can differentiate these algorithms according to three criteria: initialization method, processing mode, and type of output. This classification is logical as it aligns with the algorithms' processing flow, including how they are initiated, processed, and, ultimately, their results.

Regarding the initialization method criteria, two main approaches depend on the detection method used. Detection-Based tracking (DBT) algorithms, as illustrated in Figure 7, require a detection algorithm, and the object trajectory is obtained afterward. Alternatively, Detection-Free tracking (DFT) requires manual detection of a certain number of objects. DBT is much more widely used because it can handle appearances and reappearances throughout the sequences. Evaluating this algorithm depends on the detection algorithm's performance. The presence of two computationally intensive components, detection and tracking, significantly increases system complexity, requiring resource optimization for real-time performance [22].

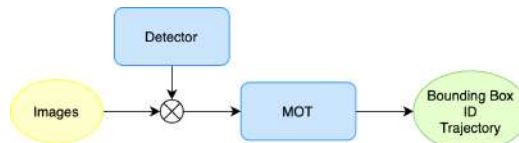


Figure 7: Representation of the DBT framework.

On the other hand, and according to the processing mode criteria, MOT algorithms are categorized as online or offline. Online tracking exclusively employs frames from the past, whereas offline tracking incorporates both past and future frames. Although offline tracking exhibits superior tracking performance, it is not viable for real-time applications.

Regarding the type of output criteria, the MOT algorithms can be classified as deterministic or probabilistic. Probabilistic tracking methods, produce varying results due to randomness, while deterministic tracking methods, produce consistent results. The primary distinction between these two methodologies lies in the optimization techniques employed.

2.4 Image Fusion

Image fusion is the process of merging complementary information from multiple source images into a single output image that is more informative and allows for superior performance in specific tasks such as detection or tracking. Given the remarkable proficiency of DL methods in image fusion, their application in ITC and RGB image fusion has gained importance [23].

Several image fusion algorithms have been proposed over the years and can be classified into pixel-level, feature-level, and decision-level fusion algorithms [24]. Conversely, a simple approach can be adopted, classifying the fusion methods into two categories: early fusion, which closely corresponds to pixel-level fusion, and late fusion, which aligns with feature-level fusion.

Early fusion combines data from multiple modalities, such as ITC and RGB images, directly at the input stage before any feature extraction or analysis, as illustrated in Figure 8. This approach integrates pixel-level information, preserving the original details and enabling models to exploit the complementary aspects of each modality. Early fusion works best with spatially aligned data, enhancing the detail of inputs for tasks like object detection or classification.

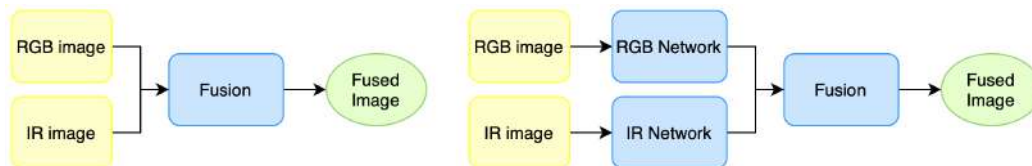


Figure 8: Early (*left*) and late (*right*) fusion.

Late fusion integrates information at the output stage after individual modalities have undergone independent processing. It combines features, intermediate representations, or final decisions from separate models to generate a unified outcome. This approach is advantageous when the modalities differ significantly in scale or characteristics, as it allows each to be processed optimally without interference. However, it often incurs higher time and computational costs because each modality must be processed independently before integrating its outputs.

2.5 Performance Evaluation

Performance evaluation of a framework and its algorithms is essential because it allows for a direct comparison with other methods. To assess our MOT framework, two aspects of performance must be evaluated: object detection accuracy and tracking performance.

To obtain the object detection performance metrics, it is essential to take into account some detection outcomes such as [25]: True Positive (TP) that corresponds to an object correctly detected, False Positive (FP) that corresponds to an object that does not exist but is identified as positive, and False Negative (FN) that corresponds to an object incorrectly detected. These detection outcomes can be used to obtain the Precision (P) that evaluates how accurate the model's positive predictions are according to:

$$P = \frac{TP}{TP + FP}. \quad (3)$$

and the Recall (R) that measures the model's ability to identify actual positive cases correctly according to:

$$R = \frac{TP}{TP + FN}. \quad (4)$$

There is often a trade-off between these metrics since increasing the number of detected objects and improving R may lead to more false positives and lowering P. To address this trade-off, the Average Precision (AP) metric uses the P-R curve, which plots precision against recall across varying confidence thresholds. The AP metric provides a comprehensive evaluation by calculating the area under this curve [26]. The AP is defined as:

$$AP@_{\alpha} = \int_0^1 P(R) dR \quad (5)$$

where $P(R)$ is P as a function of R, and the integration is performed over the R values ranging from 0 to 1. The parameter α specifies an Intersection over Union (IoU) (6) threshold that determines whether a detected object is considered as TP.

When analyzing BBs, there is often a slight discrepancy in their location and size between the detection result and the Ground Truth (GT). The IoU metric quantifies this discrepancy by calculating the ratio of the overlapping area between the predicted BB and the GT BB to the total area of their

union according to:

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (6)$$

Detection metrics focus on location and classification, while tracking performance necessitates a different approach. The main goal in tracking is to maintain object identities over time, assessing the existence of Identity Switches (IDS), which introduces challenges like ensuring consistency of identities and handling occlusions. Thus, tracking metrics must evaluate the accuracy of BB predictions and their temporal consistency across frames.

Multi-Object Tracking Accuracy (MOTA) is a metric commonly used to assess the accuracy of MOT algorithms and penalizes FP and FN detections. MOTA quantifies the number of times a GT trajectory is interrupted during the tracking process as follows:

$$\text{MOTA} = 1 - \frac{\sum_{t=1}^T (\text{FN}_t + \text{FP}_t + \text{IDS}_t)}{\sum_{t=1}^T \text{GT}_t}, \quad -\infty \leq \text{MOTA} \leq 1 \quad (7)$$

where T corresponds to the total number of frames in the evaluation sequence, FN_t represents the number of FN, FP_t the number of FP, IDS_t the number of IDS, and GT_t the total number of GT objects in frame t .

Higher Order Tracking Accuracy (HOTA) is a metric that evaluates tracking performance by balancing detection and association accuracy, providing a more nuanced assessment than MOTA [27], calculated as:

$$\text{HOTA}_\alpha = \sqrt{\frac{\sum_{c \in \{\text{TP}\}} \mathcal{A}(c)}{|\text{TP}| + |\text{FN}| + |\text{FP}|}} \quad (8)$$

where α specifies a IoU (6) threshold that determines whether a detected object is considered as TP, c is a particular individual true positive instance from the set $\{\text{TP}\}$, and $\mathcal{A}(c)$ is defined as:

$$\mathcal{A}(c) = \frac{|\text{TPA}(c)|}{|\text{TPA}(c)| + |\text{FNA}(c)| + |\text{FPA}(c)|} \quad (9)$$

where $\text{TPA}(c)$ are the true positive associations, $\text{FNA}(c)$ are the false negative association, and $\text{FPA}(c)$ are the false positive associations as described in more detail in [27]. The HOTA metric ranges from 0 to 1, where higher values indicate better overall tracking performance.

3 State of the Art

Reviewing current SOTA implementations is essential when developing a new approach. Understanding the existing aerial person detection and tracking methods and the datasets available for research is necessary. Table 1 summarizes research on person detection and tracking relevant to this thesis, highlighting algorithms, data collection sensors, and results.

Article	Algorithm	Application	Sensors	Results
Speth, et al. (2022) [10]	YOLOv3	Detection	RGB & ITC	97.9 AP@0.5
Lygouras & Gasteratos (2020) [6]	F-RCNN	Tracking	RGB & ITC	Not given
Banuls, et al. (2020) [11]	YOLOv3	Detection	RGB & ITC	88.7 AP@0.5
Bachir & Memon (2024) [28]	YOLOv5	Detection	RGB	96.9 AP@0.5
Sambolek & Ivasic-Kos (2021) [7]	YOLOv4	Detection	RGB	96.0 AP@0.5
Martinez-Alpiste, et al. (2021)[8]	YOLOv3	Detection	RGB	39.9 AP@0.5
Dousai & Loncaric (2021) [29]	EfficientDET	Detection	RGB	93.3 AP
Jiang, et al. (2022) [18]	YOLOv3	Detection	ITC	73.9 AP
Dong, et al. (2021) [9]	YOLOv3	Detection	ITC	85.3 AP

Table 1: Overview of the most relevant research in the study field.

A significant aspect of this thesis involves the development of a novel dataset that represents missing persons from the UAV perspective using RGB and ITC images. Table 2 summarizes existing datasets for person detection, highlighting each dataset name, data type, and accessibility status. Availability as Open Source (OS) is crucial for model training or validation with other datasets. All described datasets are OS, essential for a diverse dataset representing the final application.

Source	Name	RGB	Thermal	RGB Res	Thermal Res	Objective
[30]	WiSARD	15 453	15 453	3840 × 2160	640 × 512	Detection
[10]	NII-CU	5 880	5 880	3840 × 2160	640 × 512	Detection
[31]	UMA-SAR	2 288	2 288	704 × 576	704 × 576	Detection
[32]	OSU 3	3 011	3 011	320 × 240	320 × 240	Detection
[33]	RGBT234	116 900	116 900	630 × 460	630 × 460	Tracking
[34]	SARD	1 981	-	1920 × 1080	-	Detection
[35]	HERIDAL	68 750	-	4000 × 3000	-	Detection
[36]	UAV123	113 476	-	1280 × 720	-	Detection
[37]	UAV Thermal	-	77 365	-	640 × 512	Detection
Proposed	CSTE	-	-	1920 × 1080	1280 × 1024	Tracking

Table 2: Overview of the most relevant datasets in the study field.

This chapter will explore in more detail the SOTA research described, focusing on person detection (Section 3.1), person tracking (Section 3.2), image fusion (Section 3.3), SAR applications (Section 3.4), datasets (Section 3.5), and in describing some of the existing literature gaps (Section 3.6).

3.1 Person Detection

Object detection represents a fundamental yet challenging task in CV. The evolution of DL networks and advancements in Graphical Processing Unit (GPU) computing power have significantly boosted the performance of detection systems, allowing the introduction of new techniques for enhancing speed, accuracy, and robustness in complex environments[4]. For instance, DL-based models such as You Only Look Once (YOLO), one-stage, and Faster-Regional Convolutional Neural Networks (F-RCNN), two-stage, employ CNN to obtain real-time performance and high accuracy. Transformer-based models, such as DETection TRansformer (DETR), utilize bipartite matching and directly predict the one-to-one object set for end-to-end object detection without needing hand-crafted anchor boxes [38]. More recent publications such as [38, 39, 40] demonstrated the effectiveness of DETR in object detection. In particular, Cooperative-DETR has achieved the top ranking in the Microsoft Common Objects in Context (COCO) benchmark [41].

The YOLO family of object detection systems has become a cornerstone in real-time detection applications. A detailed analysis of the evolution of YOLO is presented in [26], spanning from its original version to some of the latest iterations, including YOLOv8 and YOLO-Neural Architecture Search (NAS) models. Although the first version of YOLO was unveiled in 2016, numerous new versions have been developed since then, each introducing enhancements in accuracy, speed, and overall performance, with one of the latest being YOLOv11 by Ultralytics [42]. As discussed in [18], the YOLO algorithm is a good candidate for detecting objects from UAVs. Specific versions, such as YOLO-Drone [43] and Drone-YOLO [44], have also demonstrated promising results.

3.2 Person Tracking

Being able to track individuals across video frames using CV is vital, especially for applications such as SAR operations. The field of MOT has advanced significantly with the rise of DL and improved GPU computational power, enabling higher accuracy and robustness. Several SOTA MOT algorithms have been proposed, each with unique strategies for tracking. Simple Online and Real-time Tracking (SORT) is one of the earliest and most widely used MOT algorithms, known for its efficiency and simplicity. It combines Kalman Filters (KFs) for motion prediction with the Hungarian algorithm

for data association [45]. However, SORT relies solely on geometric information, which can lead to identity switches in complex scenes. To address this limitation, Deep Simple Online and Real-time Tracking (DeepSORT) introduces appearance modeling using a DL-based feature extractor, which significantly improves tracking performance in crowded and dynamic environments [46]. This approach reduces IDS by matching detections based on geometry and appearance similarity.

A recent advancement, ByteTrack, enhances robustness by incorporating high-confidence and low-confidence detections into the data association process [47]. This strategy minimizes missed detections and significantly reduces IDS, making ByteTrack particularly effective in scenarios with occlusions or overlapping objects. Numerous studies have shown the practical efficacy of MOT algorithms. For instance, [22] developed a near-real-time tracking system that attained a MOTA score of 62.1, making it competitive in real-time contexts.

3.3 Image Fusion

Due to the complementary features provided by visible and Infrared (IR) images, a comprehensive review of RGB-IR fusion algorithms was presented in [24]. DL methods achieve state-of-the-art performance, while graph-based and correlation methods present inferior but competitive results. A review of image fusion algorithms based on Neural Networks (NNs) is presented in [23]. The review evaluates qualitative and quantitative fifteen fusion methods where DenseFusion, Proximity Maximization Gradient Integration (PMGI), and Structure-Preservation Detail Network (SDNet) demonstrated a good overall performance while exhibiting the most efficient processing times.

A perspective on image fusion in DL is presented in [48], introducing several applications for the algorithms, including RGB-IR for object tracking. It is noted that IR images substantially highlight persons from the background, producing images with high contrast and rich textures and highlighting the relevance of this image fusion step in this study.

3.4 Search And Rescue (SAR) Applications

For detecting persons in need in disaster scenarios, [10] created a new dataset with UAV RGB and ITC images. Four approaches were investigated, with the most relevant one employing a four-channel that combines

RGB and thermal images to use one image with 4 channels as input. The other approach was an ensemble model that combines two YOLO networks in the two types of information. Overall, the four-channel model with the YOLOv3 algorithm performed better, achieving an AP@0.5 of 97.9. A closer examination of the dataset substantiates such a high average accuracy. The dataset's videos exhibit a remarkable simplicity, as they consistently feature individuals in the same environment with a consistent background.

Using ITC and RGB images for automated object detection prompted the creation of two novel datasets by [11]. These datasets were generated utilizing an all-terrain vehicle for SAR operations equipped with an ITC and a RGB camera. In this study, YOLOv3 was employed in both datasets, achieving an AP@0.5 of 87.4 for the RGB dataset and 88.4 for the thermal dataset in the class *Person*.

These two works [10, 11] are the most similar to our work, combining information from RGB with ITC images in SAR environments. As described in Table 3, we can see that the combination of the two types of information, although barely, obtained a better result than the two types separately. It is common in both works that information from ITC yields better results.

Model	[10]	[11]
RGB (only)	92.4	87.4
ITC (only)	97.5	88.4
Ensemble	97.3	-
4-Channel	97.9	-

Table 3: Ap@0.5 of different models used in person detection.

3.5 Datasets

In the domain of person detection, several datasets are readily available. However, the selection becomes more limited regarding aerial point-of-view datasets, with only a handful of datasets available in SAR environments. Furthermore, the only datasets found that meet the criteria of containing synchronous RGB and ITC images were the National Institute of Informatics - Chiba University (NII-CU), UMA-SAR, and RBGT234 datasets [10, 31, 33].

The NII-CU dataset [10] comprises 5,880 pairs of aligned RGB and ITC images captured from a UAV operating at altitudes between 20 and 50 me-

ters, with the cameras positioned at a 45-degree downward angle. A homography warping technique was employed to rectify lens distortion and align the ITC images with the RGB images. The individuals in the frames were annotated with rectangular BBs.

The UMA-SAR Dataset [31] comprises a comprehensive collection of multimodal raw data acquired from a manned all-terrain vehicle during two realistic outdoor SAR exercises conducted by actual emergency responders. The sensor suite includes an RGB camera and a ITC, giving a total of 2,288 pairs of images. The mission's primary objective was to gather diverse data from the SAR domain, encompassing individuals, vehicles, debris, and SAR activity on uninhabited terrain. The images were annotated with rectangular BBs.

The RBGT234 [33] dataset was crafted to tracking purposes, leveraging the synergistic benefits of thermal information in conjunction with visible data. This dataset stands out with its extensive size, encompassing 233.8k frames, facilitating large-scale performance evaluation. Moreover, the alignment between RGB and ITC image pairs is exact, eliminating the necessity for pre and post-processing. Rectangular BBs were utilized for annotation.

3.6 Literature Gaps

This chapter reviewed the most recent advancements in person detection in SAR environments. The most recent research in this field is limited to specific datasets or does not address real-time applications.

Regarding the existing datasets, it is evident that most papers misrepresent the expected real SAR environment. While it is true that most datasets are small, they often contain information in similar environments. For instance, a field with grass or a footpath alone cannot be considered a SAR environment. Furthermore, most datasets consist of people only walking without the necessary realism. In a SAR environment, it is crucial to consider scenarios where individuals lie on the ground in various positions. Additionally, partial occlusion of a person must be considered, mainly when accounting for the overlapping, e.g., of low vegetation, trees, and rocks.

Lastly, time is decisive in SAR operations. Therefore, these algorithms' real-time performance and accuracy must be highlighted to ensure their feasibility during operations. A more tailored dataset will bring challenges, making it essential to ensure the correct adaptations and the use of SOTA algorithms to achieve a proper trade-off between speed and accuracy.

4 Methodology

Based on the previously described, exploring the methodology that will be employed during the thesis development is essential. For that purpose, an overview of the proposed approach will be detailed (Section 4.1), the creation of the dataset (Section 4.2), the algorithms (Section 4.3), and the training process, since we will mainly use DL methods (Section 4.4).

4.1 Overview

We propose three approaches based on the data type. The first approach utilizes only RGB data, as illustrated in Figure 9, while the second approach utilizes only ITC data, as shown in Figure 10. The third approach utilizes both RGB and ITC data by creating new frames using a fusion algorithm, as illustrated in Figure 11.

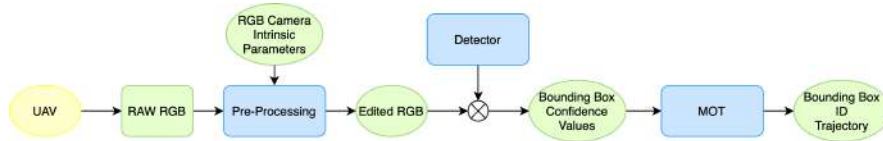


Figure 9: RGB based framework.

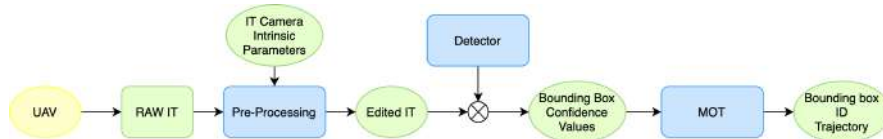


Figure 10: ITC based framework.

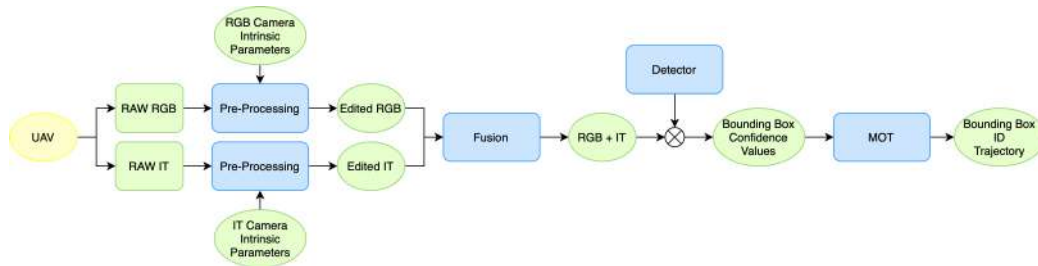


Figure 11: Fusion based framework.

All approaches start by acquiring UAV data, requiring independent pre-processing of various input types. This involves standardizing frame rates and resolutions and using the camera's intrinsic parameters to produce undistorted images. This is a crucial step to ensure a good alignment of frames captured by different cameras. Once the pre-processing is complete, the edited frames serve as input to the detection model in both the first and second approaches, Figure 9 and 10, respectively. The detection algorithm identifies the person's location, represented by a BB, and assigns a confidence value to the detection. After successful detection, the MOT algorithm assigns a unique ID to the detected BB and tracks its movement across frames.

The third approach integrates RGB and ITC images by fusing both data types. The fusion combines these data types as input to generate a new frame that encapsulates relevant information from each modality. The resulting frame is then utilized as input for the detection and tracking algorithms, similar to approaches 1 and 2, as illustrated in Figure 11.

The three proposed approaches aim to explore the effectiveness of different data modalities in SAR scenarios. The first two approaches utilize individual data types, focusing on simplicity and computational efficiency. These methods are particularly suited for scenarios where visible light (daytime) or thermal imaging (low light/nighttime) provides sufficient information. The third approach merges RGB and ITC images to combine complementary data. This method improves robustness and accuracy by utilizing each type's strengths and adapting to challenging lighting and occlusion.

4.2 Dataset Creation

Besides being a clear contribution to the field, creating and proposing a new dataset in this thesis are among the fundamental components of the work's development. As mentioned in Section 1, we will collaborate with EPRU, a team specializing in SAR missing person operations utilizing UAVs. This highly specialized and trained team is exclusively dedicated to finding missing individuals. Their collaboration will enhance the utility of this work, as everything will be adapted wherever possible to real-world scenarios.

The EPRU team will handle video acquisition and recording during operations. All recorded videos should depict real-life disappearance scenarios. Due to data sensitivity, the dataset cannot be publicly shared without permission. If the videos lack diversity in backgrounds, altitudes, or positions, we will complement the dataset with videos of missing persons in varied but

controlled simulated environments.

As mentioned in Section 1, all videos will be acquired using the DJI Matrice 30T [5] equipped with a gimbal, which features zoom and wide-angle cameras, as well as an ITC. The RGB camera records video at 3840×2160 pixels at 30 frames per second (fps), but 1920×1080 pixels at 60 fps is preferred due to smaller file sizes. In contrast, the ITC has a lower resolution of 1280×1024 pixels at 30 fps. However, it is still superior to the camera resolutions in the datasets described in Section 3.

Currently, we possess approximately 12 minutes of RGB/ITC videos that represent approximately 21 thousand frames of each data type. All the videos need preprocessing to identify relevant sequences. In addition to this, an essential aspect is annotating the frames. We will use annotation tools like Computer Vision Annotation Tool (CVAT), a leading engine for machine learning image and video annotation by Intel [49]. *Labelbox* is another platform for annotating images and videos for machine learning training, used by companies such as OpenAI and Google [50].

4.3 Algorithms

As outlined in Section 3, several algorithms can be used for object detection, object tracking, and image fusion. Based on the current SOTA, the chosen one-stage detector is YOLO, and the chosen transformer-based detector is DETR (Section 4.3.1). Subsequently, after evaluating the performance of both detectors, tracking algorithms such as SORT and ByteTrack will be employed, as they are among the most promising options (Section 4.3.2). Concerning image fusion, the most promising algorithms are DenseFuse, PMGI, and SDNet (Section 4.3.3).

4.3.1 Detection

The YOLO algorithm, introduced by [51], revolutionized object detection and has become the most popular framework for real-time object detection. Unlike conventional methods that adapt classifiers for detection, YOLO re-defined object detection as a regression task. This entails predicting spatially separated bounding boxes and their corresponding class probabilities within a unified framework. Using NNs, YOLO processes entire images in a single evaluation, directly outputting bounding box coordinates and class predictions. The seamless integration of the detection pipeline into a single network

facilitates end-to-end optimization, enhancing detection performance.

DETR is a novel object detection framework that introduces transformers into the detection pipeline, eliminating the need for traditional hand-crafted components like anchor boxes and non-maximum suppression. By framing object detection as a direct set prediction task, DETR employs an encoder-decoder transformer architecture to model global relationships between objects and the image. It processes an input image through a CNN backbone to extract features, which are then passed to the transformer. Using a fixed number of learnable queries, the transformer simultaneously predicts object bounding boxes and class labels, making it a one-stage detector. This end-to-end approach simplifies training and inference while achieving competitive accuracy, particularly for scenes with complex object interactions [52].

4.3.2 Tracking

SORT is a widely used MOT algorithm that prioritizes efficiency and simplicity. SORT combines Kalman filters for state estimation with the Hungarian algorithm for frame-by-frame association of detected objects. It assumes that object detections are provided as input and maintains a separate track for each object, updating positions and velocities over time. Despite its simplicity, SORT performs remarkably well in real-time applications, particularly when minimal occlusions or identity switches occur. However, its reliance on only geometric information for association can lead to identity switches in crowded or complex scenes, limiting its performance in highly dynamic environments [45].

ByteTrack, on the other hand, is a more recent MOT algorithm that builds on the simplicity of SORT while addressing its limitations. ByteTrack introduces a robust association strategy that considers high-confidence and low-confidence detections, enabling the tracker to recover from missed detections or occlusions. By incorporating low-confidence detections as potential candidates for matching, ByteTrack significantly reduces identity switches and improves overall tracking accuracy. This makes it particularly effective in crowded scenes and challenging real-world scenarios [47].

4.3.3 Fusion

DenseFusion is an image fusion algorithm that uses a dense convolutional network architecture. By incorporating dense connections into its encoder,

it connects the outputs of all preceding convolutional layers as inputs to subsequent layers, enhancing feature extraction capabilities and mitigating information loss in deeper network layers. This architecture ensures the effective preservation of structural details and thermal information during pixel-level (early) fusion.

PMGI focuses on gradient-based fusion, ensuring that structural and textural information is maximally preserved from both input images. PMGI achieves visually appealing results with clear boundaries by optimizing gradient proximity, effectively enhancing image clarity in complex scenes.

The SDNet emphasizes preserving structural details while enhancing fine textures during image fusion. It operates at the feature level, extracting modality-specific features and fusing them into a unified representation. SDNet is particularly effective in scenarios requiring high-quality fused outputs, such as medical imaging and security applications.

4.4 Training

Given the substantial time and resource requirements for training a model from scratch, pre-trained DL models will be utilized, focusing on fine-tuning whenever possible. An example of this is the YOLO version from Ultralytics, mentioned in Section 3.1, which is already pre-trained on the Microsoft dataset COCO. An illustration of the pre-trained YOLO model in a real captured SAR video sequence is shown in 5.3.

Dividing the dataset into three sets is standard practice when training a DL model. This division ensures the model has adequate training, evaluation, and generalization performance. A standard dataset split involves three sets: a training set containing 70-80% of the data, a validation set containing 10-15% of the data, and a test set containing 10-15% of the data. The training set allows the model to learn patterns and adjust the model's parameters. The validation set serves as a platform for fine-tuning hyperparameters and detecting overfitting. Finally, the test set evaluates the model's performance and generalization capabilities on unseen data.

Another commonly used technique is cross-validation, which involves dividing the dataset into multiple subsets or folds. The model is trained and validated on different combinations of these folds, ensuring that each fold eventually serves as training and validation data. This approach provides a more robust evaluation by reducing the variance introduced by a single train-validation split.

5 Preliminary Results

To conduct a preliminary exploration of the methods with the hardware that will be used during the development of this work, calibration of both RGB camera and ITC was performed to correctly correlate the obtained data (Section 5.1). Following this, a preliminary fusion of RGB and ITC images is performed, highlighting the impact of preprocessing on the fusion process (Section 5.2). Finally, the YOLO detection algorithm is applied to a few sample frames from both modalities, providing an initial assessment of the detection capabilities in the proposed workflow (Section 5.3).

5.1 Cameras calibration

The camera calibration process was performed using a *Python* Open Source Computer Vision Library (OpenCV) implementation [53] to ensure accurate alignment and correction of distortions for both the RGB and ITC images. This involved capturing multiple frames containing a 9×6 chessboard pattern to extract corresponding object points, as illustrated in Figure 12. The calibration procedure uses these points to compute the camera intrinsic matrices and distortion coefficients described in Section 2.1, enabling the correction of radial and tangential distortions.

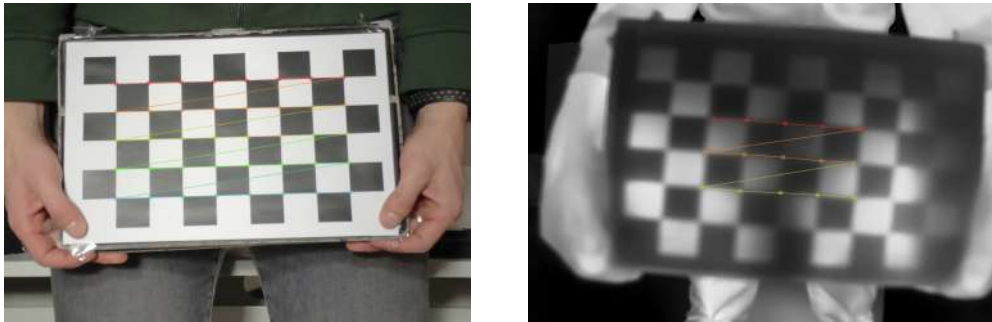


Figure 12: RGB camera calibration (*left*) and ITC calibration (*right*).

For the RGB camera, we have obtained the following intrinsic matrix:

$$\mathbf{K}_{\text{RGB}} = \begin{bmatrix} 2825.431 & 0.0 & 933.139 \\ 0.0 & 2811.541 & 507.293 \\ 0.0 & 0.0 & 1.0 \end{bmatrix} \quad (10)$$

with the radial and tangential distortion coefficients $[K_1, K_2, p_1, p_2] = [0.255, -0.631, 0.001, -0.002]$. On the other hand, for the ITC, we have obtained the following intrinsic matrix:

$$\mathbf{K}_{\text{ITC}} = \begin{bmatrix} 10717.918 & 0.0 & 612.585 \\ 0.0 & 9839.016 & 586.215 \\ 0.0 & 0.0 & 1.0 \end{bmatrix} \quad (11)$$

with the radial and tangential distortion coefficients $[K_1, K_2, p_1, p_2] = [-26.267, 3550.381, -0.037, 0.072]$. Figures 13 and 14 illustrate examples of distorted and undistorted RGB and ITC images, respectively.



Figure 13: RGB distorted frame (*left*) and undistorted frame (*right*).



Figure 14: ITC distorted frame (*left*) and undistorted frame (*right*).

Reprojection error evaluates calibration accuracy. It indicates the precision of parameters, with lower values signifying better accuracy. Zero pixels indicate perfect calibration, while values below one pixel are satisfactory. An

error of one pixel means the calculated image point is offset by one pixel from its actual position. For the RGB camera, we achieved a reprojection average error of 0.08 pixels, and for the ITC, we achieved a reprojection average error of 1.04 pixels. Both obtained errors indicate satisfactory calibration results. Conversely, the result for the ITC is significantly larger than that for the RGB camera. While this result is acceptable, a more refined approach to camera calibration is necessary during the thesis.

The RGB frame transitions from a resolution of 1920×1080 to 1914×1064 , and the ITC frame undergoes a more significant reduction from 1280×1024 to 1220×976 . These changes are caused by cropping during the undistortion process. Although the impact on RGB resolution is more negligible, 0.31%, the 4.69% reduction in ITC resolution may affect data fusion, highlighting the need for further refinement in pre-processing.

5.2 Image Fusion

Despite the inherent imperfections in the ITC calibration, the fusion results demonstrate a notable enhancement in integrating information from both RGB and ITC. However, while the fusion effectively captures complementary features, such as thermal and visual details, it falls short of being excellent. Minor misalignments and noise introduced by the calibration inaccuracies limit the overall quality, suggesting the need for further refinement to achieve optimal results. Figure 15, where the fusion process is performed using SDNet, demonstrates this, where certain image regions exhibit a distinct misalignment between the two data types. Conversely, specific segments exhibit a satisfactory overlay of the objects, making this approach promising.

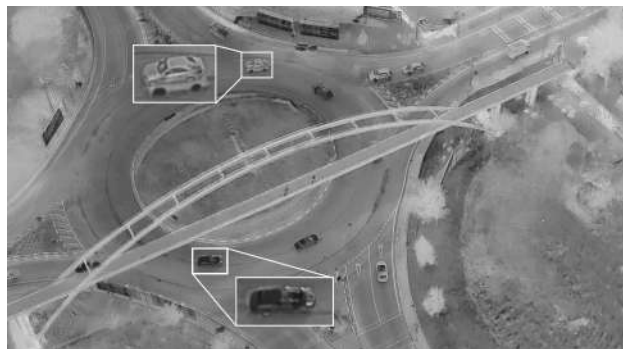


Figure 15: Image fusion with SDNet.

5.3 Detecting algorithms

The YOLOv8 model from Ultralytics was tested on both the RGB and ITC frames to evaluate its initial performance without training it in our dataset. Despite being trained on the COCO dataset, the model demonstrated the ability to detect objects in both modalities, showcasing the robustness of pre-trained weights across domains. However, the detection was notably poor in both data types, highlighting the need for additional training on modality-specific data, a process known as fine-tuning. As expected, the model performed better on the RGB data compared to the ITC data, likely due to the closer similarity of RGB data to the images in the pre-training dataset. As shown in Figure 16 and 17, the model predominantly detected cars in the RGB and ITC datasets, while detecting people proved challenging at high UAV altitudes, with successful detections occurring only at lower altitudes. These results underline the potential of transfer learning while emphasizing the importance of domain adaptation for optimal performance.



Figure 16: YOLOv8 in RGB data (*left*) and ITC data (*right*).



Figure 17: YOLOv8 in SDNet fusion data.

6 Work Plan

In this chapter, we outline our work plan for this thesis. We detail the tasks (Section 6.1) and the expected execution timeline (Section 6.2), describing potential risks and providing possible solutions (Section 6.3).

6.1 Tasks

As expected, the tasks to be performed during this thesis strongly align with the secondary objectives described in Section 1.3.

1. **Acquisition of a New Dataset:**
 - (a) Data collection;
 - (b) Data organization.
2. **Dataset Preprocessing:**
 - (a) Data cleaning;
 - (b) Image fusion;
 - (c) Data annotation.
3. **Implementation and Testing of Algorithms:**
 - (a) Training with RGB camera data;
 - (b) Training with ITC data;
 - (c) Training with fusion data;
 - (d) Evaluation of algorithm performance.
4. **Real-Word Testing:**
 - (a) Field Deployment;
 - (b) Performance Monitoring;
 - (c) Feedback and Iteration.
5. **Proposed Architecture Optimization:**
 - (a) Model optimization;
 - (b) Validation of Performance Trade-offs;
 - (c) Obtaining the best model.
6. **Thesis Writing:**
 - (a) Restructure, correction, and adaption of the thesis proposal;
 - (b) Improve the SOTA, background and methodologies description;
 - (c) Results from Analysis and Training using the created dataset;
 - (d) Results from real-world tests and architecture optimization;
 - (e) Writing the final model performance evaluation and conclusions.

6.2 Timeline

The preliminary work plan illustrated in Figure 18 is adjusted to the thesis development period and will be adapted as needed.

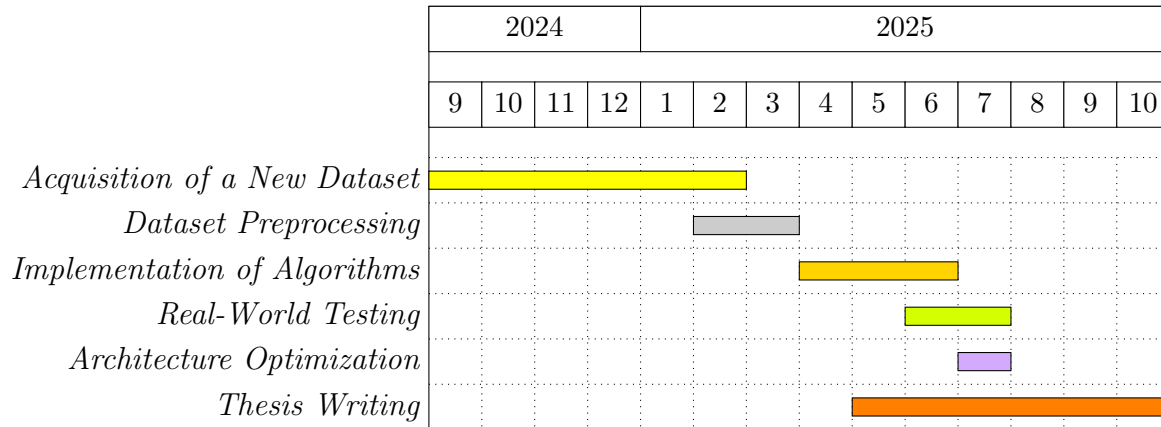


Figure 18: Preliminary work plan.

6.3 Risks

Most of the risks were mitigated mainly due to the collaboration of EPRU, as this collaboration will ensure access to hardware, real SAR video sequences, and operational procedures essential to the development of this thesis. The risks identified so far are described in Table 4.

Description	Solutions	Risk
Objects too small	Increase resolution Image enhancement Decrease UAV altitude	Medium
RGB and ITC images not sufficiently aligned	Late fusion 4-channel detector	Low
Previous studies have shown better results	Model optimization	Low

Table 4: Description of the risks, solutions, and their degrees of importance.

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